

Logistic Equations Effectively Model Mammalian Cell Batch and Fed-Batch Kinetics by Logically Constraining the Fit

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A four-parameter logistic equation was used to fit batch and fed-batch time profiles of viable cell density in order to estimate net growth rates from the inoculation through the cell death phase. Reduced three-parameter forms were used for nutrient uptake and metabolite/product formation rate calculations. These logistic equations constrained the fits to expected general concentration trends, either increasing followed by decreasing (four-parameter) or monotonic (three-parameter). The applicability of this approach was first verified for Chinese hamster ovary (CHO) cells cultivated in 15-L batch bioreactors. Cell density, metabolite, and nutrient concentrations were monitored over time and used to estimate the logistic parameters by nonlinear least squares. The logistic models fit the experimental data well, supporting the validity of this approach. Further evidence to this effect was obtained by applying the technique to three previously published batch studies for baby hamster kidney (BHK) and hybridoma cells in bioreactors ranging from 100 mL to 300 L. In 27 of the 30 batch data sets examined, the logistic models provided a statistically superior description of the experimental data than polynomial fitting. Two fed-batch experiments with hybridoma and CHO cells in benchtop bioreactors were also examined, and the logistic fits provided good representations of the experimental data in all 25 data sets. From a computational standpoint, this approach was simpler than classical approaches involving Monod-type kinetics. Since the logistic equations were analytically differentiable, specific rates could be readily estimated. Overall, the advantages of the logistic modeling approach should make it an attractive option for effectively estimating specific rates from batch and fed-batch cultures.

Introduction

There has been an increasing use of mammalian cell cultures for the manufacture of complex therapeutic proteins. However, protein yields from mammalian culture are relatively low, requiring the optimization of cell lines, medium formulations, and bioprocesses. These optimization efforts typically first involve evaluating noninstrumented batch cultivations (typically <100 mL working volume) in T-flasks, spinners, or roller bottles where cell growth, metabolism, and protein productivity are monitored over the course of the experiment. This provides information used to select the cell lines and medium components that maximize protein yields. Further bioprocess optimization, in either fed-batch or continuous perfusion cultivations, is mainly performed using laboratory-scale (>1 L) bioreactors. Although maximizing specific protein productivity is often the primary objective in laboratory-scale experiments, ensuring robust cell growth and metabolism are also important. All of these variables of interest are quantified using cell specific rates that enable comparison of cell lines and

cultivation conditions. Accurate estimation of specific rates is thus vital to meaningfully interpret results from bioprocess optimization experiments.

Whereas specific rates for steady state perfusion cultures are readily computed because of their relatively time-invariant nature, analyzing the dynamic kinetics of batch and fed-batch cultures is more challenging. A conventional approach to model mammalian cells in batch and fed-batch cultures has been through the use of unstructured kinetic models, variations of the classical Monod equation (1–5), and reviews of these models are available (6, 7). Although unstructured kinetic models have adequately described experimental data, they are computationally not practical to implement as they involve nonlinear estimation of a large number of kinetic parameters from a system of differential equations. Moreover, given the variety of unstructured kinetic models that have been used to describe mammalian cell cultures, comparisons between studies is complicated. Analytical solutions of the differential equations describing the state variables have also been used to estimate specific rates (8–11). These solutions, however, are derived under the assumption that the specific rates are constant, as can be expected during the exponential growth phase. These have limited applicability to the other phases of batch and fed-batch cultures where

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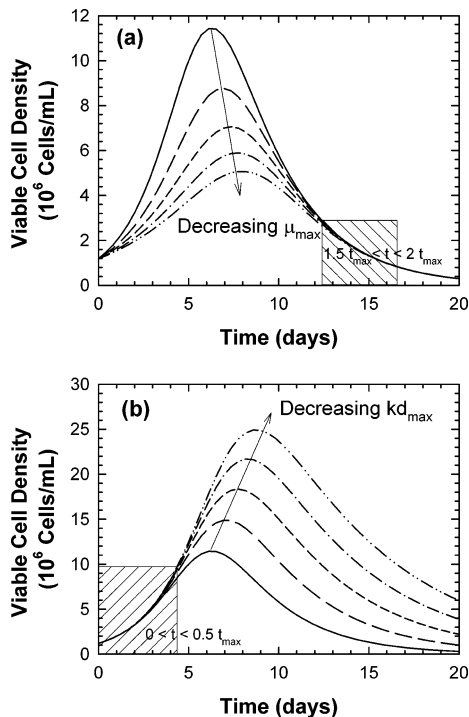


Figure 1. Sensitivity of the viable cell density curve to the logistic parameters $D(\mu_{\max})$ and $B(k_{d_{\max}})$. Successive curves are for 25% decreased parameters compared to the previous curve.

specific rates are not constant. Specific rates in fed-batch cultures have also been estimated from the slope on plots of cumulative state variables (nutrient, metabolite or product) versus integral viable cell density (12–13). This approach provides an average estimate of the specific rate of interest over the exponential growth phase, but additional linear or nonlinear fits need to be used for other cultivation phases. This need for multiple fits to describe the time course of a single variable makes this approach cumbersome and prone to error.

A general approach that is applicable over the entire time course of cultures is fitting polynomials to the data (14, 15). This approach has been used for both batch and fed-batch cultivations and is attractive because it allows simplified computation of the time derivatives necessary to estimate cell specific rates. However, as time profiles of cellular, nutrient, and product concentrations exhibit exponential behavior, they are difficult to describe by polynomials (16). For instance, two polynomial functions were necessary to describe the time course of some state variables (14). Moreover, polynomial fits are known to yield unrealistic trends, especially when the data include even a few outliers.

Logistic equations have been successfully used to describe population dynamics in a variety of applications (17–23) but have not been reported to model experimental data from mammalian cell batch or fed-batch cultures. Most reported applications involve bacterial growth curves characterized by lag, exponential, and stationary phases that are adequately described by the logistic growth equation. Mammalian cells in batch and fed-batch cultivations also exhibit a sharp decline in cell density following the stationary phase, a behavior that cannot be described by the standard logistic growth equation. In addition, a decline in lactate concentration during later stages of fed-batch cultivation is also frequently observed (13, 24). Alternate logistic formulations that incorporate both the ascending and descending components of cell growth are available (25).

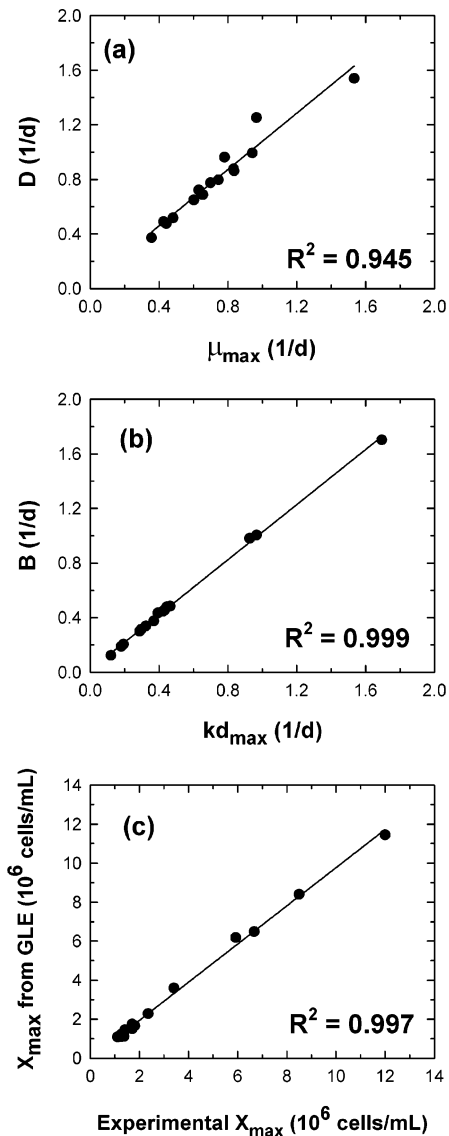


Figure 2. Illustration of the biological significance of the logistic parameters using eight batch and seven fed-batch cell density data sets (1, 14, 33, 34).

This study presents a method for modeling batch and fed-batch mammalian cell culture data using logistic equations. An alternate logistic formulation was applied to cases where variables had both increasing and decreasing phases. Cell specific rates were readily obtained from the analytically differentiable logistic equations. A comparison was made between this logistic modeling approach and the polynomial fitting or the unstructured kinetic modeling approaches that are commonly used to describe batch and fed-batch data.

Theory

Calculation of Batch Culture Specific Rates. Most batch kinetic studies have used discrete forms to compute specific rates. The widespread use of this approach is primarily due to its simplicity as seen from the specific growth rate expression

$$\mu' = (\mu - k_D) = \frac{\Delta X_v}{\bar{X}_v \Delta t} \quad (1)$$

where μ' is the apparent specific growth rate (1/day) over an interval from t_1 to t_2 , μ is the actual specific

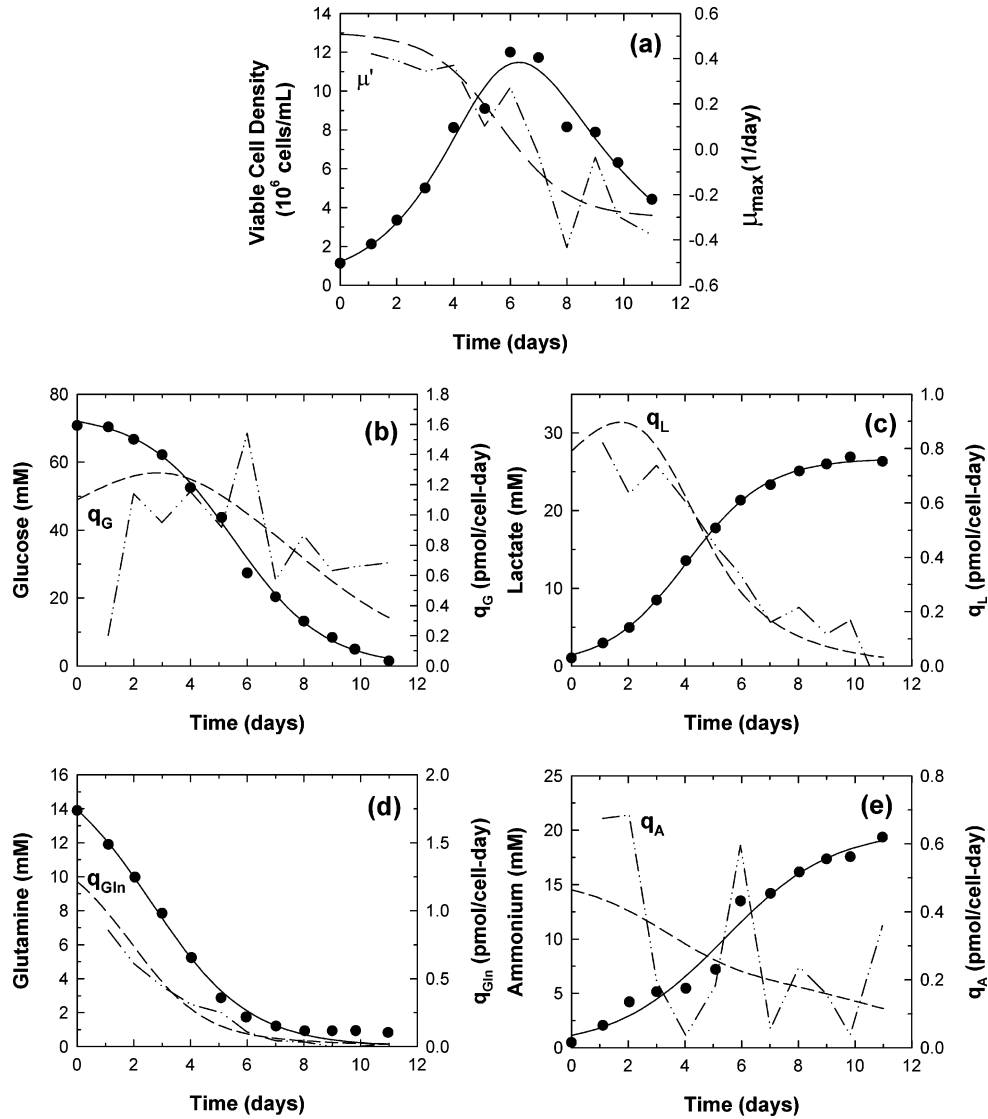


Figure 3. Time profiles of cell density, nutrient, and metabolite concentrations for CHO cells in 15-L batch culture. Experimental data (●); logistic (GLE for cell density, LDE for glucose and glutamine, and LGE for lactate and ammonium) fit (—); logistic specific rate (---); discrete derivative specific rate (- · - ·).

growth rate (1/day), k_D is the specific death rate (1/day), X_v is the viable cell density ($\times 10^6$ cells/mL), $\Delta X_v = X_{v2} - X_{v1}$, t is time (day), $\Delta t = t_2 - t_1$, and $\bar{X}_v = [X_{v2} + X_{v1}]/2$. Thus ΔX_v and Δt represent the difference between successive viable cell density and time points, respectively, while $\bar{X}_v$ is the arithmetic average of the consecutive cell density data points. A log-normal average can also be used for $\bar{X}_v$ during the exponential growth phase, but this provides a poor estimate for the average X_v in the decline phase. For intervals of constant apparent growth rate, a more accurate estimate of μ' can be obtained by combining multiple data points. However, when this constant growth rate ends and how the growth rate changes beyond that point remain difficult to accurately compute.

The primary sources of carbon and energy in a typical mammalian cell culture medium are glucose and glutamine. The specific consumption rate of glucose can be calculated from

$$q_G = -\frac{\Delta G}{\bar{X}_v \Delta t} \quad (2)$$

where q_G is the specific glucose consumption rate

(pmol/cell-day) and $\Delta G = G_2 - G_1$ is the difference in glucose concentration over consecutive data points (mol/L). The primary known toxic metabolites of glucose and glutamine metabolism are lactate and ammonium, respectively. These metabolite as well as protein production rates can be calculated from

$$q_P = \frac{\Delta P}{\bar{X}_v \Delta t} \quad (3)$$

where q_P is the specific production rate (pg/cell-day) and $\Delta P = P_2 - P_1$ is the change in metabolite/product concentration over consecutive data points (g/L).

Calculation of Fed-Batch Culture Specific Rates. Fed-batch cultivations typically involve the periodic feeding of glucose, glutamine, and other medium components. Hence, nutrient mass balance expressions are modified to take the feeding into account, whereas the expressions for cell density, metabolites, and product are essentially identical to those in a simple batch cultivation (when dilution effects can be neglected). For example, specific glutamine uptake rate in a continuously fed batch reactor can be described by

$$q_{Gln} = -\frac{\Delta Gln}{\bar{X}_v \Delta t} + \frac{F}{V} \frac{Gln_f}{\bar{X}_v} - k_{Gln} \frac{Gln}{\bar{X}_v} \quad (4)$$

where q_{Gln} is the specific glutamine uptake rate (pmol/cell-day), $\Delta Gln = Gln_2 - Gln_1$, the change in glutamine concentration over consecutive data points (mol/L), F is the glutamine feed rate (L/day), V is the bioreactor working volume (L), Gln_f is the glutamine concentration in the feed (mol/L), and k_{Gln} is the first-order abiotic degradation constant for glutamine with values depending on the medium composition and culture pH (26).

General Equation Describing the Dynamics of Batch and Fed-Batch Cultures. The selection of final process parameters from optimization experiments is derived mainly from comparisons of cell specific productivity and growth rate. Therefore it is important to reliably estimate these rates and this can be done from sequential data points and eqs 1–4. However, this typically yields erratic time profiles since this method is sensitive to the measurement errors common in biological systems. Mathematical models that describe the dynamics of cellular growth and protein production according to expected trends can provide more robust estimates of the cell specific variables of interest. Such models should more smoothly and logically fit the experimental data. For instance, to describe viable cell density (X) in batch and fed-batch systems requires initially increasing and subsequently decreasing levels. This cannot be fit by a simple exponential growth model (eq 1) or by the commonly used logistic growth equation. It is proposed that a four-parameter generalized logistic equation (GLE) should be used to describe viable cell concentration:

$$X = \frac{A}{\exp(Bt) + C \exp(-Dt)} \quad (5)$$

where A , B , C , and D are non-negative model parameters (25). It is informative to relate the logistic equation parameters to corresponding biological process parameters. The contribution of $\exp(Bt)$ is minimal in the growth phase and when set to zero, eq 5 reduces to an exponential growth equation with D as the specific growth rate. Similarly, neglecting contributions of $C \exp(-Dt)$ during the cell death phase reduces eq 5 to an exponential decay expression with B as the death rate. The parameters D and B thus represent the maximum growth rate, μ_{max} , and the maximum death rate, $k_{d,max}$, respectively. These would be constant in the exponential growth and corresponding decline phases, respectively. Simulations were performed to test this hypothesis and Figure 1 illustrates sensitivity of the cell density curve to D and B and hence μ_{max} and $k_{d,max}$ values. As expected, changes to μ_{max} affect the exponential growth phase, whereas the influence is negligible in the decay phase, especially for $t > 1.5t_{max}$ (Figure 1a). Sensitivity to $k_{d,max}$ is minimal for $t < 0.5t_{max}$, whereas later portions of the cell density curve are significantly affected (Figure 1b). Equation 5 can thus be written in terms of μ_{max} and $k_{d,max}$ as

$$X = \frac{A}{\exp(k_{d,max} t) + C \exp(-\mu_{max} t)} \quad (6)$$

The initial cell density, X_0 , can be expressed in terms of A and C by setting $t = 0$ in eq 6:

$$X_0 = \frac{A}{1+C} \quad (7)$$

Table 1. Previously Published Batch and Fed-Batch Studies Used To Test the Logistic Modeling Approach Presented in This Study

ref	cell line	bioreactor type	modeling approach
1	hybridoma	stirred tank (300-L batch)	kinetic modeling
33	hybridoma	T-flask (100-mL batch)	kinetic modeling
14	BHK	spinner (500-mL batch)	polynomial fitting
15	hybridoma	benchtop (2.4-L fed-batch)	polynomial fitting
34	CHO, hybridoma	benchtop (0.7-L fed-batch)	discrete derivatives

Setting the derivative of eq 6, $dX/dt = 0$ provides an equation for t_{max} , the time corresponding to the maximum cell density:

$$t_{max} = \frac{1}{k_{d,max} + \mu_{max}} \ln \left(\frac{C\mu_{max}}{k_{d,max}} \right) \quad (8)$$

An estimate for X_{max} , the maximum cell density attained during the cultivation, can now be obtained from substituting eq 8 into eq 6:

$$X_{max} = \frac{Ak_{d,max}}{C\mu_{max} \left\{ \exp \left(\frac{k_{d,max}}{k_{d,max} + \mu_{max}} \right) + C \exp \left(-\frac{\mu_{max}}{k_{d,max} + \mu_{max}} \right) \right\}} \quad (9)$$

It must be noted that eq 5 could also fit the successive ascending and descending lactate concentrations often observed in fed-batch culture. Most other product and nutrient concentrations can be expected to monotonically increase or decrease, respectively, over the whole duration of the run (expect at times of fed-batch additions). This suggests that simplified forms of eq 5 could effectively describe the concentrations of nutrients and products in batch or fed-batch cultivations. Setting $B \rightarrow 0$ in eq 5 results in the logistic growth equation (LGE) that can be used to describe monotonically increasing product concentration, P :

$$P = \frac{A}{1 + C \exp(-Dt)} \quad (10)$$

The parameter D is a rate constant for concentration increase, and definitions of A and C can be obtained by setting $dx/dt = 0$ and $t = 0$ in eq 10, respectively

$$A = P_{max}; \quad C = \frac{P_{max} - P_0}{P_0} \quad (11)$$

where P_{max} is the maximum value of P and P_0 the initial value at $t = 0$. Using these definitions, the LGE can be rewritten to be consistent with other presentations in the literature (27):

$$P = \frac{P_0 P_{max}}{P_0 + (P_{max} - P_0) \exp(-Dt)} \quad (12)$$

Setting $D \rightarrow 0$ in eq 5 results in the logistic decline equation (LDE) that can be used to describe monotonically decreasing nutrient concentration, N :

$$N = \frac{A}{\exp(Bt) + C} \quad (13)$$

where B is a rate constant for concentration decrease and

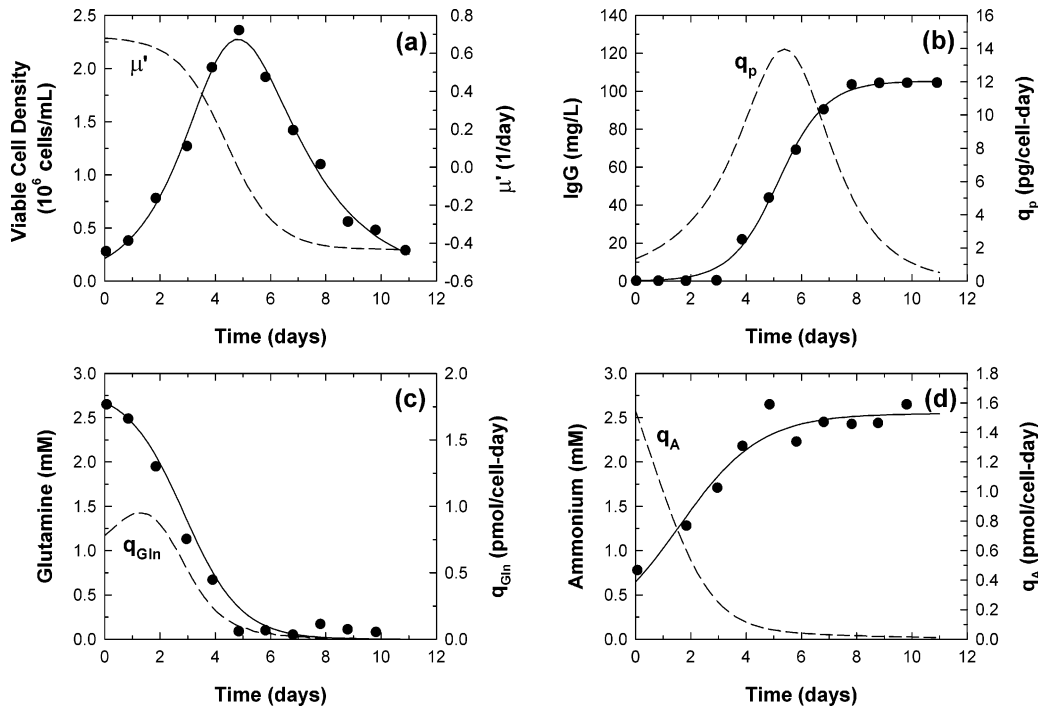


Figure 4. Viable cell density, IgG, glutamine, and ammonium concentrations for hybridoma cells in 300 L batch culture (1). The points are experimental data, and the solid lines are fits by the logistic equations (GLE for cell density, LDE for glutamine, and LGE for IgG and ammonium). Specific rates calculated from the logistic fits are shown as dashed lines.

definitions for A and C are

$$A = \frac{N_{\max} N_0}{N_{\max} - N_0}; \quad C = \frac{N_0}{N_{\max} - N_0} \quad (14)$$

with $N_{\max}$ and N_0 the maximum and initial nutrient concentrations, respectively.

Specific rates could be readily estimated from the logistic models as eqs 5, 10, and 13 are analytically differentiable:

$$\frac{dX}{dt} = X \left\{ \frac{DC \exp(-Dt) - B \exp(Bt)}{\exp(Bt) + C \exp(-Dt)} \right\} \quad (15)$$

$$\frac{dP}{dt} = PD \left(1 - \frac{P}{A} \right) \quad (16)$$

$$\frac{dN}{dt} = -NB \left(1 - \frac{C}{A} N \right) \quad (17)$$

Materials and Methods

Cell Line, Medium, and Cell Culture System. CHO cells were cultivated in batch mode with glucose and glutamine as the main carbon and energy sources in a proprietary medium formulation. Experiments were conducted in three 15-L bioreactors (Applikon, Foster City, CA) with a 10-L working volume. The temperature was maintained at 36.5 °C and the agitation at 40 rpm. The dissolved oxygen (DO) concentration was maintained at 50% air saturation by sparging a mixture of oxygen and nitrogen (100–150 mL/min) through 0.5 μ m spargers. The bioreactor was inoculated at an initial cell density of approximately 1.0×10^6 cells/mL, and the pH was maintained at 6.8 by the automatic addition of 0.3 M NaOH.

Analytical Methods. Samples from the bioreactor were taken daily for cell density and viability analysis

using the CEDEX system (Innovatis, Bielefeld, Germany). The samples were subsequently centrifuged (Beckman Coulter, Fullerton, CA), and the supernatants were analyzed for nutrient and metabolite concentrations. Glucose, lactate, glutamine, and glutamate concentrations were determined using a YSI model 2700 analyzer (Yellow Springs Instruments, Yellow Springs, OH) while ammonium was measured by an Ektachem DT60 analyzer (Eastman Kodak, Rochester, NY). The pH and DO were measured online using retractable electrodes (Mettler-Toledo Inc., Columbus, OH), and their measurement accuracy was verified through off-line analysis in a Stat Profile 9 blood gas analyzer (Nova Biomedical, Waltham, MA). The same instrument also measured the dissolved CO₂ concentration. On-line measurements of cell density were made with a retractable optical density probe (Aquasant Messtechnik, Bubendorf, Switzerland), calibrated with hemocytometer counts of cell concentrations.

Nonlinear Parameter Estimation. The parameters A , B , C , and D in eqs 5, 10, and 13 were estimated by minimizing the sum of squares error (SSE) between the experimental and model fit data:

$$\text{SSE} = \sum_{i=1}^j \{ (x_{\text{meas}})_i - (x_{\text{fit}})_i \}^2 \quad (18)$$

where $(x_{\text{meas}})_i$ is the i th experimental x value and $(x_{\text{fit}})_i$ is the i th model fitted x value in a total of j observations. Equation 5 involved the sum of exponentials and was inherently unstable. Hence three different algorithms were used for nonlinear parameter estimation, the Levenberg–Marquardt method (28), the simplex approach (29), and the generalized reduced gradient method (30, 31). The parameters used were those that resulted in the lowest values of the SSE defined in eq 18. The parameter standard errors and the correlation between parameters were estimated from the covariance matrix to help evaluate the quality of the model fit to the experimental

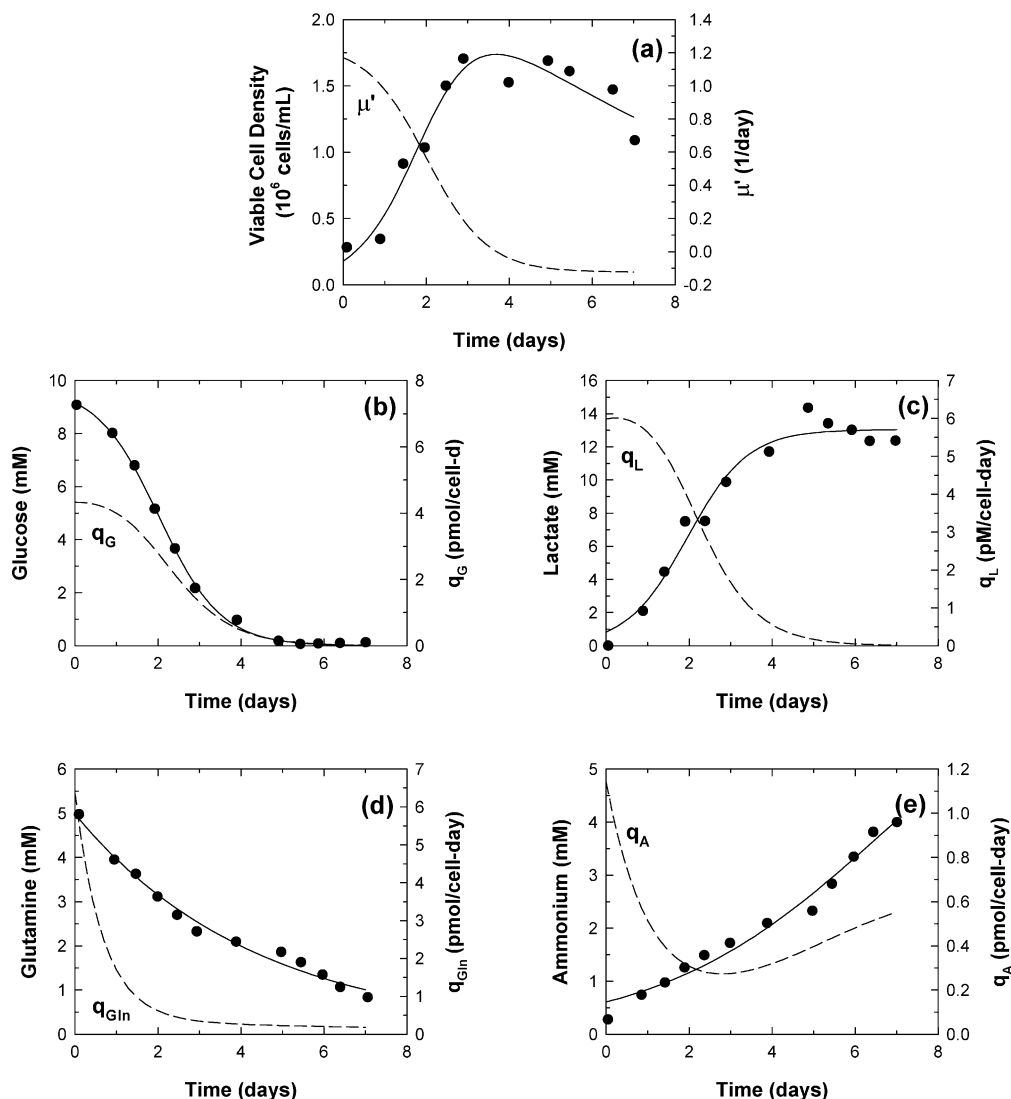


Figure 5. Viable cell density, nutrient, and metabolite concentrations for BHK cells in 500-mL batch culture (14). The points are experimental data, and the solid lines are fits by the logistic equations (GLE for cell density, LDE for glucose and glutamine, and LGE for lactate and ammonium). Specific rates calculated from the logistic fits are shown as dashed lines.

data. When multiple models with different degrees of freedom were fit to the same data set, the F test (32) was used to discriminate among the models.

Results and Discussion

Biological Significance of the Logistic Parameters. The exponential growth and death phases were defined as $0 < t < 0.5t_{\max}$ and $1.5t_{\max} < t < 2t_{\max}$, respectively, based on an examination of the cell density profiles from eq 5 (Figure 1). To verify these definitions, $\mu_{\max}$ and $k_{d\max}$ were computed from all cell concentration data analyzed in this work and compared with the logistic parameters D and B . Excellent agreement between the maximum rates and the logistic parameters was seen in all cases (Figure 2a and b) supporting the reformulation of eq 5 as eq 6. The utility of eq 9 to predict the maximum cell density in batch and fed-batch cultures was verified by comparing $X_{\max}$ values calculated from eq 9 with experimental data (Figure 2c). For all 15 data sets, experimentally observed maximum cell densities were accurately predicted by the $X_{\max}$ definition in eq 9 and the fitted logistic parameters.

Description of Experimental Data from Batch Cultures. The time profiles of CHO cell density, nutri-

ent, and metabolite concentrations along with the logistic model fits are shown in Figure 3. The GLE was first used to describe all of the state variables measured in this experiment. Subsequently, the LDE was used to describe the monotonically decreasing glucose and glutamine concentrations, while the LGE was used to describe the increasing lactate and ammonium concentrations. Model discrimination using the F test indicated that the LDE and LGE fits were statistically superior to the GLE at the 95% confidence level for the nutrient and metabolite concentrations, respectively, and hence results from these equations are presented in Figure 3. The experimental data were well fitted by the models, and the corresponding specific rates were calculated from eqs 1–3 using analytical derivatives of the logistic equations (eqs 15–17).

It is remarkable that the model fit the data so well even though logistic models do not include independent terms for growth-related and maintenance-related metabolism. This could be due in part to the predominant effect of exponential cell growth compared to relatively gradual shifts over a batch culture of growth- or maintenance-related metabolic rates. From a practical standpoint, the use of a single equation and its reduced forms to describe all experimental measurements in batch (or

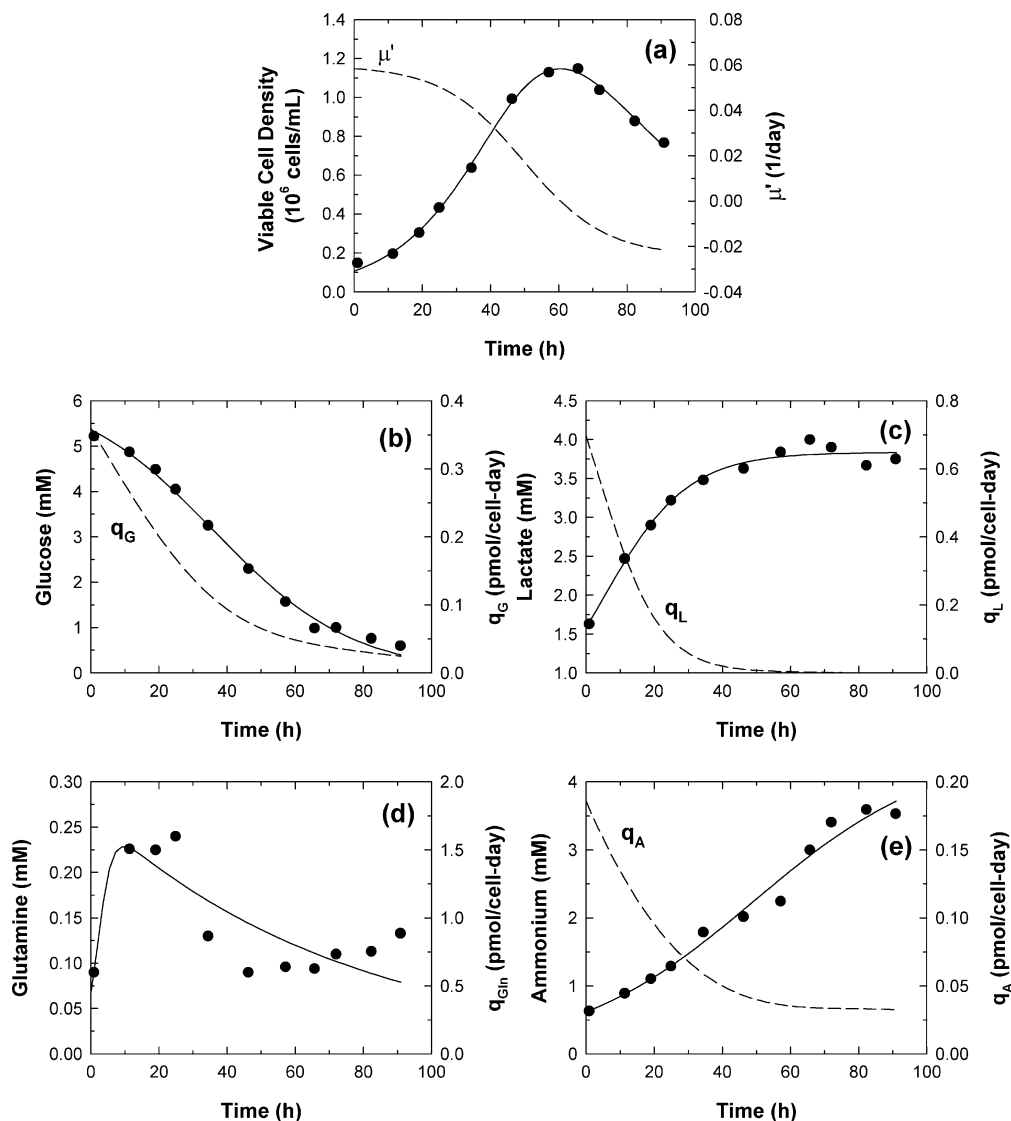


Figure 6. Viable cell density, nutrient, and metabolite concentrations for hybridoma cells in glutamine-limited 2.4-L fed-batch culture (15). The points are experimental data, and the solid lines are fits by the logistic equations (GLE for cell density, LDE for glucose and glutamine, and LGE for lactate and ammonium). Specific rates calculated from the logistic fits are shown as dashed lines.

fed-batch) cultivations adds to the simplicity of the proposed logistic approach.

Experimental data from batch cultivations are obtained from periodic samples whose concentrations are analyzed, and then the data is converted to the corresponding derivatives to obtain specific rates. The logistic equations provided smooth and close fits to all of the concentration data, thereby yielding smooth logistic specific rate profiles (Figure 3). In contrast, specific rates obtained using discrete derivatives of the state variables were not smooth and were highly sensitive to outliers in experimental measurements. Though the discrete derived q_{Gln} values were acceptable, those for glucose and ammonium were in gross error, primarily due to outliers in the experimental data. The μ' and specific lactate production rate, q_L , profiles were similar to those from the logistic fits, albeit not as smooth. In all cases besides q_{Gln} , physiologically implausible oscillations were introduced by the discrete fit and not by the logistic fit.

The more general utility of this logistic approach was further evaluated using data from published batch studies that investigated different cell lines (hybridoma and baby hamster kidney) in bioreactors ranging from 150-cm³ T-flasks to 300-L stirred tanks (Table 1). Two of these

studies (1, 33) used Monod-type kinetics to describe the experimental data while experimental data were fitted by polynomial functions in the third (14). Results from using the logistic equations to describe the data from two of these experiments are shown in Figures 4 and 5. The nutrient, metabolite, and product concentrations were fit by the LDE and LGE models, while the four-parameter GLE was necessary to describe the dynamics of viable cell concentration. In all cases, the experimental data were well described by the logistic equations and similar good fits were obtained for data from the third study (not shown). These results clearly indicate the applicability of the logistic models to describe experimental data obtained by multiple groups from batch reactors of varying sizes and cell types.

Description of Experimental Data from Fed-Batch Cultures. These logistic methods would be much more useful if they could be applied to fed-batch cultures that become the focus of later stages of development and manufacturing. However, it was a concern that periodic feeding of nutrients would distort the resulting profiles so that the logistic approach might not be suitable. Fed-batch data from two studies (15, 34) (Table 1) were analyzed using the logistic equations. In the first study

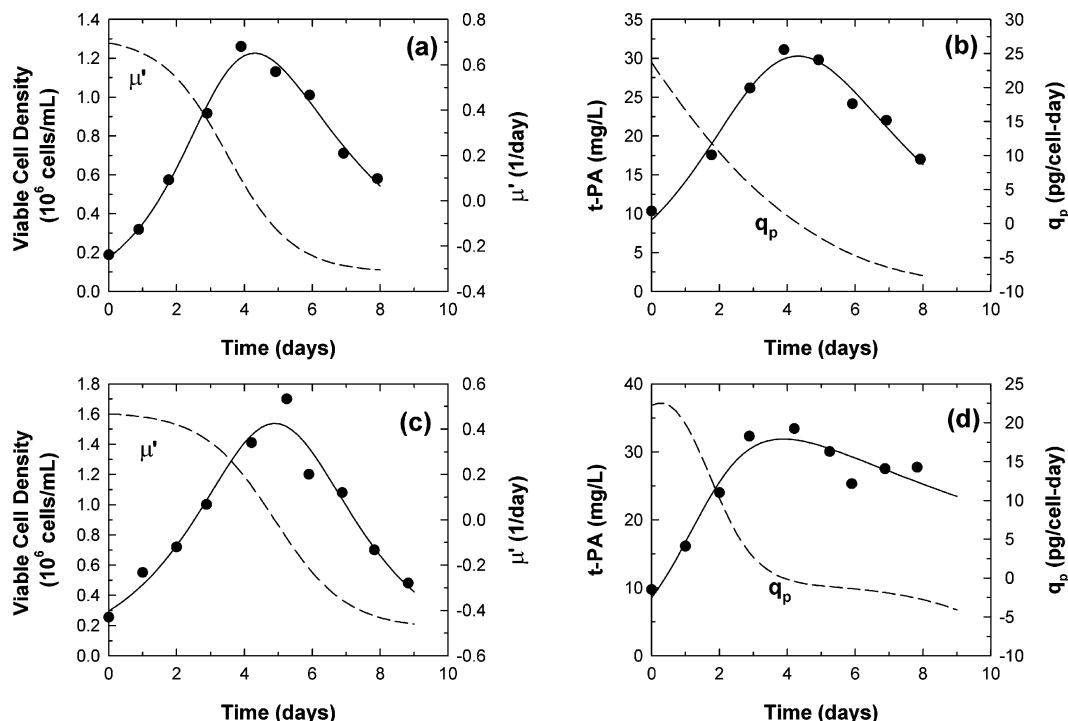


Figure 7. Viable cell density and t-PA concentration for CHO cells in 0.7-L fed-batch culture under two different feeding conditions (34). Glucose was fed at 4 pmol/cell-day for panels a and b, whereas amino acids were also fed for panels c and d. The points are experimental data, and the solid lines are fits by the logistic equations (GLE for both cell density and t-PA). Specific rates calculated from the logistic fits are shown as dashed lines.

(15), hybridoma cells were cultivated in a 2.4-L bioreactor with the feeding of glucose or glutamine or both. The second involved cultivation of tissue plasminogen activator (t-PA)-producing CHO cells in a 0.7-L bioreactor with glucose or amino acid feeding. Data from glutamine-limited fed-batch hybridoma cultures are shown in Figure 6 along with corresponding logistic fits. All variables except glutamine (the nutrient that was fed) were fit well by the logistic equations. Time profiles of the fed nutrient will depend strongly on the feeding strategy, often with concentrations at low values to minimize the production of metabolites (13, 35). The logistic equations cannot be expected to effectively fit such fed-nutrient profiles. A total of 20 data sets were analyzed from the CHO fed-batch cultivations (34), and representative cell density and t-PA concentration data under two different feeding conditions are shown in Figure 7. The logistic equations fit the data well, as was true for the remaining 16 data sets (not shown). It should be noted that the four-parameter GLE was used to describe t-PA concentration because of the declining trend later in the culture. Similar declines have been observed for lactate concentration in fed-batch cultures (24), and in such instances, the GLE (as opposed to the LGE) most effectively fits that experimental data.

Comparison with Other Modeling Approaches. Polynomial approximation and unstructured kinetic modeling are the primary methods currently used to fit data from batch and fed-batch experiments. Given the conceptual similarity between the logistic modeling approach presented in this study and polynomial approximation, it is important to compare their ability to describe mammalian cell culture data. Polynomial approximation has limitations because exponential state variable time profiles are difficult to describe with polynomial functions (16). The same 30 data sets from batch cultures were also analyzed using polynomial fitting, and the inability to describe the experimental data was quantified by the eq

18 sum of squares errors to compare with logistic fitting. Since increasing the order of a polynomial function could result in a better fit to experimental data, polynomial functions with one additional parameter compared to the corresponding logistic equation were also evaluated. For instance, the viable cell density description using eq 5 has four parameters, and this was compared with polynomials of orders 3 (4 parameters) and 4 (5 parameters). Comparisons between logistic equations and polynomials of the same order used the SSE values while comparison between logistic equations and higher order polynomials was done using the *F*-test. The *F*-test determined if the higher order polynomial fit was indeed a closer representation at the 95% confidence level.

Of the 30 batch data sets examined using both the logistic and polynomial approaches, the polynomial approach was statistically superior in only three instances. It is important to note that even in the few cases of statistical superiority for the polynomial fit these were not necessarily improved specific rate estimates. Figure 8 shows specific glutamine consumption rates for CHO cells that was one of the above-mentioned three cases where the polynomial fit was statistically superior to the logistic fit. An examination of the specific rate data from the polynomial fit indicates that data after $t = 9$ days were negative, suggesting net glutamine production. This is not reflective of the biology and is an artifact due to an inflection in the polynomial fit to the data at $t = 9$ days and beyond. The logistic modeling approach does not suffer from such errors. Instead the fits are constrained to the expected trends, monotonic in this case.

Logistic and polynomial fits to the data from Dalili et al. (33) are shown in Figure 9 as examples of the data sets where the logistic approach was statistically superior. Both polynomials grossly misrepresented the time course of viable cell density, and no improvement was obtained by increasing the order of the polynomial. Computation of growth rates from either of the poly-

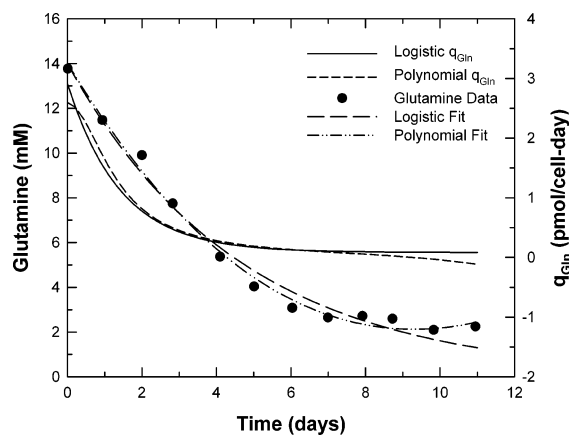


Figure 8. Comparison of q_{Gln} values from logistic (LDE) and polynomial fits for CHO cells in 15-L batch culture. The polynomial fit to glutamine depletion data was statistically superior than the logistic fit for this data set.

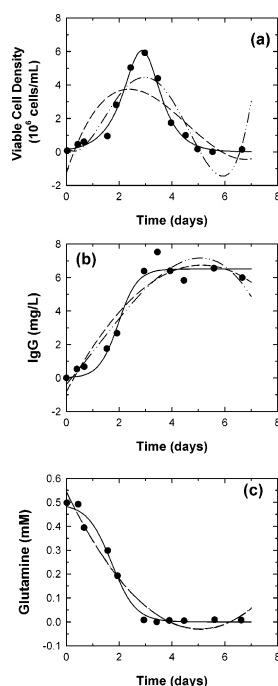


Figure 9. Comparison of logistic (GLE for cell density, LGE for IgG, and LDE for glutamine) and polynomial fits for batch cultivation of hybridoma cells in 100-mL T-flasks (33). (—) logistic fit; (---) polynomial fit with the same number of parameters as the logistic fit; (- · - ·) polynomial fit with one additional parameter (The two polynomial fits in panel c overlap).

mial fits would not be acceptable. Similar limitations, albeit to a lesser extent, were seen for the IgG and glutamine data sets in Figure 9. The polynomial approach thus lacks generality and cannot be relied upon as a robust tool for specific rate estimation in batch cultures.

Unstructured kinetic modeling involves using Monod-type equations and estimating a large number of kinetic parameters by nonlinear optimization in the system of differential equations. Two batch studies (1, 33), whose data was used in this study for verification of the logistic approach, employed modified forms of the Monod equation in kinetic expressions to describe their experimental data. One used simpler kinetic expressions for glutamine-limited cultures (33), requiring 8 kinetic parameters to be estimated, whereas 13 kinetic parameters were required for a more comprehensive kinetic model (1). Estimating such a large number of kinetic parameters

through nonlinear least squares from a system of nonlinear differential equations is not trivial and is unlikely to provide robust parameter estimates as a result of the strong correlation among the kinetic parameters. Moreover, the large variability in the kinetic models used (6, 7) makes it difficult to compare results from different studies employing this approach.

Computation of Integral Quantities. The integral viable cell density is an important parameter for the characterization of batch and fed-batch cultures (36). Since it corresponds to the area under the curve in a plot of viable cell density versus time, it can be computed by integrating eq 5. However, eq 5 cannot be analytically integrated because of the sum of exponential terms in the denominator, and an approximation to this integral is presented in the Appendix. Alternatively, a simple numerical technique, such as the trapezoidal rule (37), can be used to estimate the integral viable cell density from eq 5.

Data for Estimation of Metabolic Fluxes. An original motivation for this work was in the context of metabolic flux analysis that is increasingly used to characterize cellular metabolism and physiology by estimating fluxes through the pathways of central carbon metabolism (38). Input data for metabolic flux analysis include specific uptake and production rates in addition to cellular growth rate. The logistic equations presented in this study provide a practical means of more reliable specific rate estimation that should enable more robust metabolic flux computation in batch and fed-batch cultures.

Conclusions

The application of logistic equations for analyzing mammalian cell batch and fed-batch data has been illustrated. Though nonmechanistic in nature, these equations did provide a means to impose logical general constraints on the fitted profiles. Simplified logistic equation forms were selected on the basis of expected monotonic or increasing followed by decreasing trends. Time profiles of cell density, nutrients, and metabolites were well fitted by the logistic equations, and time derivatives of these variables were readily computed, resulting in rapid estimation of specific rates. Besides providing valuable information on cellular physiology and metabolism, specific rates are precursors for metabolic flux estimation, thereby allowing improved use of information collected in batch and fed-batch cultivations. This functional representation also allowed for computation of integral viable cell density, an indicator of batch and fed-batch process performance. Another advantage of the logistic approach is its general nature, which increases its applicability to a wide variety of experimental systems as shown in this study. This general nature coupled with the ability to rapidly obtain more robust specific rate estimates should make it an attractive alternative for describing the dynamics of mammalian cell growth and protein production in batch and fed-batch culture. This philosophy of empirical modeling based on constraining fits to expected trends could be extended to the derivation of other useful models where the complexity of systems makes mechanistic models impractical.

Appendix

An approximation to the integral viable cell density can be obtained by integrating eq 5

$$\int X dt = \frac{Ae^{Dt}}{CD} H(t) \quad (19)$$

where

$$H(t) = F\left\{1, \frac{D}{B+D}, \frac{B+2D}{B+D}, -\frac{e^{(B+D)t}}{C}\right\} \quad (20)$$

is a hypergeometric function of the form $F\{a, b, c, z\}$ -with the series expansion

$$F\{a, b, c, z\} = \sum_{k=0}^{\infty} \frac{(a)_k (b)_k z^k}{(c)_k k!} \quad (21)$$

References and Notes

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